KLU + PATH3 · ML-01 · QUICK-COMMERCE DELIVERY INTELLIGENCE

DeliveryIQ

Business Insights Report — Business Solution Document

AI-Powered Quick Commerce Delivery Intelligence System · 31 October 2026

Team	Genesis Grid
Group	Group 4

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Source: `reports/business_insights_report.md` (unchanged). Late = delivery > 30 minutes is a project-defined benchmark, not an official SLA.

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DeliveryIQ — Business Insights Report

AI-Powered Quick Commerce Delivery Intelligence System · Stage 5 · generated 2026-10-31 by `src/business_insights.py`
Every number is computed from project artifacts and stored in `reports/stage5_metrics.json`. **Labels:** [Observed] measured · [Action] recommendation · [Scenario] hypothetical, not measured.

1. Executive Summary

- **Data:** 45,593 delivery orders across 22 Indian cities, 2022-02-11 → 2022-04-06. All rows were kept; data-quality problems are flagged, not deleted.
- **ETA prediction at order placement:** test MAE **6.19 min** vs 7.47 for a median guess, **17.1% better**. R^2 0.310.
- **Late-order risk (> 30 min, project benchmark, not an official SLA):**
  - ROC-AUC **0.791**; catches **81.3%** of late orders; **46.7%** of alerts are correct (1.63× the base rate).
  - The riskiest 10% of orders were 69.8% late.
- **What drives delay (association, not proven cause):** time of day first, then delivery distance, festival days and city type. Restaurant location, calendar and order type add almost nothing.
- **When to surge-staff:** orders placed 19:00–21:59 are about half late, against 29.8% for the network (§4.1). The three evidence-backed findings are in §4.6.
- **Geography:** the 22 clusters are the 22 cities (silhouette 0.948 = city separation). Late rates do not differ significantly between them (p = 0.103). **No dark-store location can be justified from this data.**
- **Rider segmentation is not feasible:** rider IDs do not identify individual riders.
- **After rider assignment** a comparison model reaches MAE 3.67 min (without rider rating). That is strong evidence for refreshing the ETA at dispatch, but the model is not production-ready.
- **Three priorities:**
  1. Deploy order-placement late-risk alerts with tiered interventions
  2. Introduce a two-stage ETA: estimate at placement, refresh at rider assignment
  3. Surge-staff riders in the 19:00–21:59 window and plan promises around predictable peaks

2. Business Problem and Objectives

Customers need a reliable delivery promise, and operations need early warning of orders that will run late. The project answers four questions: how long will an order take, will it be late, where is demand, and what should management do.

Objective	Approach	Status
Predict delivery ETA	Gradient-boosting regression vs median baseline, ridge, random forest	✅ MAE 6.19 min
Predict late-delivery risk	Gradient-boosting classifier; > 30 / 35 / 40 min benchmarks	✅ ROC-AUC 0.791
Geographic delivery zones	K-Means with elbow + silhouette	⚠️ Done; zones = cities, so limited operational value
Rider segmentation	—	❌ Not feasible with the available rider IDs (§9)
Three business recommendations	Evidence from Stages 1–5	✅ §10
Interactive demo	Streamlit	⏳ Stage 6

3. Dataset and Data Quality Summary

Item	Value	Handling
Orders	45,593 (20 raw columns)	All 45,593 retained
Exact duplicate rows	0	—
Rows with ≥ 1 missing value (stored as text `NaN`)	4,225	Converted to real blanks; never filled in the data
Restaurant coordinates with a flipped minus sign	431	Corrected only after 4 geographic checks; originals kept
Restaurant location recorded as (0, 0)	3,640	Flagged invalid; coordinates blanked, never invented
Placeholder rider records (age 15 / rating 1, age 50 / rating 6)	91	Age and rating blanked; originals kept
Missing order time	1,731	Kept as missing; never rebuilt from pickup time
Missing festival flag	228	Kept; no 'is-missing' feature (it marks an artifact)

Food delivery, not quick commerce: order types are Snack, Meal, Drinks and Buffet from restaurants. The median delivery takes 26 minutes over a median 9.2 km. Findings describe restaurant delivery and transfer to grocery quick commerce only by analogy.

Synthetic patterns: delivery points are diagonal offsets from restaurants (98.2% of rows); delivery times cluster in 5-minute blocks; traffic level follows order hour almost exactly (97.48%); several segments are late 100% of the time (§4). Real-world performance will differ.

4. Key EDA Findings

Overall: mean delivery 26.3 min, median 26; **29.8% of orders exceed 30 minutes**. These are observed associations across all orders.

Factor	Group	Orders	Mean ETA (min)	% late > 30
Order time	Morning 08–10	5,756	19.5	2.2
Order time	Midday 11–14	4,429	26.9	32.0
Order time	Afternoon 15–18	10,340	26.7	32.5
Order time	Evening 19–23	22,907	27.8	35.5
Order time	Night 00 (only hour 0 in data)	430	22.2	9.8
Order time	(order time missing)	1,731	26.3	30.2
Distance	0–5 km	11,303	22.2	12.4
Distance	5–10 km	11,307	24.4	21.1
Distance	10–15 km	11,293	29.8	45.5
Distance	15–25 km	8,050	29.9	45.1
Distance	(invalid location)	3,640	26.1	28.8
Festival	No	44,469	26.0	28.6
Festival	Yes	896	45.5	100.0
Festival	(missing)	228	11.2	0.0
City type	Urban	10,136	23.0	18.7
City type	Metropolitan	34,093	27.3	33.3

Factor	Group	Orders	Mean ETA (min)	% late > 30
City type	Semi-Urban	164	49.7	100.0
City type	(missing)	1,200	22.1	15.4
Traffic*	Low	15,477	21.3	7.9
Traffic*	Medium	10,947	26.7	32.5
Traffic*	High	4,425	27.2	33.6
Traffic*	Jam	14,143	31.2	50.6
Traffic*	(missing)	601	26.5	30.9
Batched deliveries*	0.0	14,095	22.9	18.1
Batched deliveries*	1.0	28,159	26.9	30.3
Batched deliveries*	2.0	1,985	40.5	100.0
Batched deliveries*	3.0	361	47.8	100.0
Batched deliveries*	(missing)	993	22.8	18.6
Order type	Snack	11,533	26.3	29.8
Order type	Meal	11,458	26.4	30.0
Order type	Drinks	11,322	26.2	29.8
Order type	Buffet	11,280	26.3	29.7

* Known only after dispatch or of uncertain timing, so excluded from the placement models.

Takeaways:

- **Time of day is the largest observed contrast:** 2.2% late in the morning vs 35.5% in the evening. The 15:00–23:59 window carries 72.9% of orders and 84.5% of late orders.
- **Distance matters, as a step:** orders over 10 km are about 46% late vs 12% under 5 km.
- **Deterministic patterns (synthetic-data warning):** every festival-day order, every Semi-Urban order and every order batched with 2–3 others is late. Real operations rarely look like this, so treat these as dataset artifacts.
- **Order type has no effect** (all about 30% late).

4.1 Late rate by hour of order (PDF visual set)

Hour	Orders	Mean ETA	% late	vs network (pp)
00	430	22.2	9.8	-20.1
08	1,818	19.6	2.8	-27.1
09	1,947	19.5	1.6	-28.3
10	1,991	19.5	2.2	-27.7
11	1,962	26.4	30.0	+0.1
12	892	26.8	32.0	+2.1
13	784	27.5	35.5	+5.6
14	791	27.5	33.8	+3.9
15	873	23.2	15.8	-14.0
16	709	22.9	14.8	-15.0
17	4,278	27.4	36.1	+6.3
18	4,480	27.3	35.1	+5.2
19	4,595	30.8	49.4	+19.6

Hour	Orders	Mean ETA	% late	vs network (pp)
20	4,539	31.2	50.6	+20.7
21	4,686	31.1	50.1	+20.3
22	4,576	23.2	14.6	-15.2
23	4,511	22.5	12.3	-17.6

Peak-risk hours: **19:00, 20:00, 21:00**. Hours above the network late rate: 11, 12, 13, 14, 17, 18, 19, 20, 21. Chart: `figures/26_late_rate_by_hour.png`.

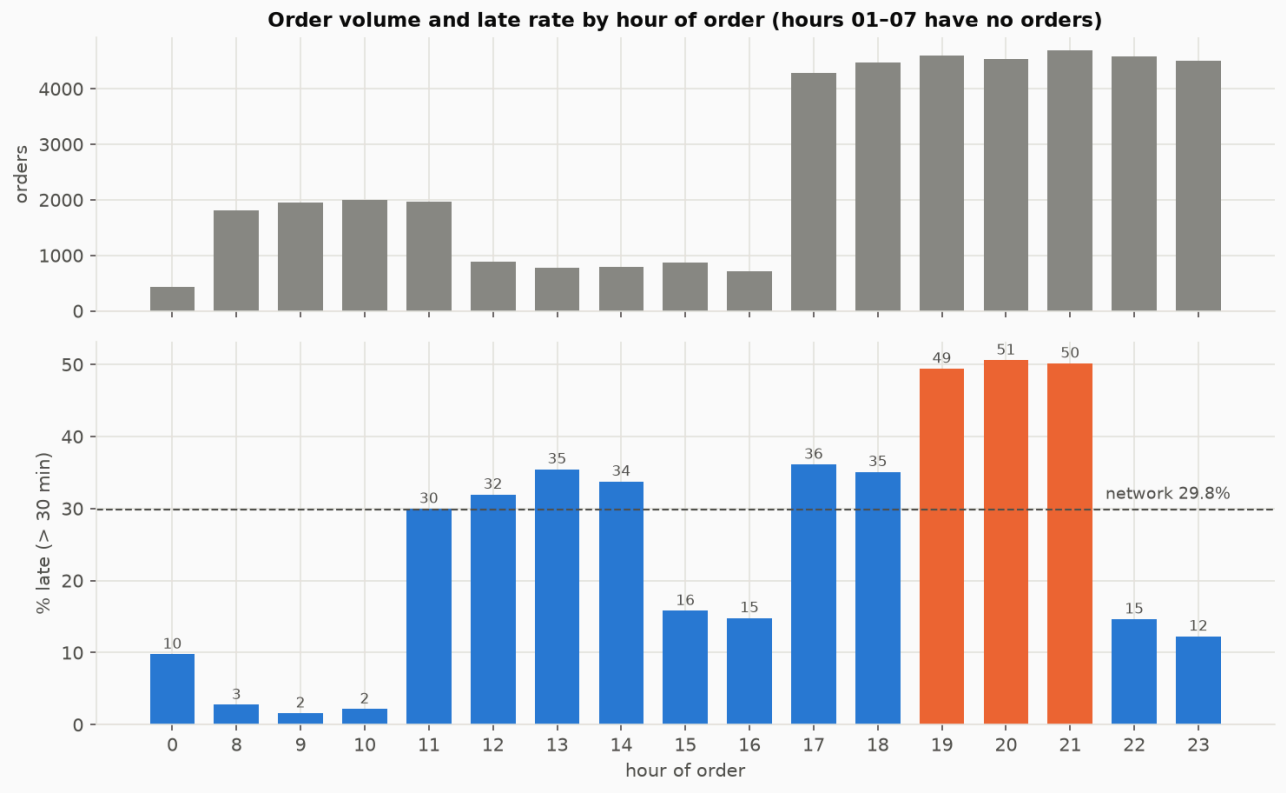


Figure 26 — Late rate by hour (source: figures/26_late_rate_by_hour.png)

4.2 Day of week

Day	Orders	Mean ETA	% late
Mon	6,207	26.3	29.5
Tue	6,375	25.3	25.5
Wed	7,093	27.8	36.2
Thu	6,348	25.2	25.2
Fri	7,031	26.8	32.2
Sat	6,290	26.1	29.0
Sun	6,249	26.4	30.2

Late rates range from 25.2% (Thu) to 36.2% (Wed); chi-square p = 1.4e-57. Caution: each weekday covers only 6–7 dates in the data, so this pattern may not repeat.

4.3 Weather vs traffic — does weather add delay beyond traffic? (PDF §8)

Weather \ Traffic	Low	Medium	High	Jam
Sunny	10.6%	6.5%	17.1%	17.8%

Weather \ Traffic	Low	Medium	High	Jam
Windy	3.0%	31.1%	34.1%	45.6%
Stormy	2.8%	31.1%	38.7%	43.4%
Sandstorms	2.0%	31.1%	38.4%	44.5%
Cloudy	14.3%	48.3%	37.6%	74.8%
Fog	14.6%	45.9%	35.3%	76.1%

Yes, mostly in heavier traffic [Observed]. Inside the same traffic level, Fog is +4.1 pp vs Sunny in Low traffic, +39.3 pp in Medium, +18.2 pp in High and +58.3 pp in Jam. Sunny orders are rarely late at any traffic level. Because traffic follows the order hour almost exactly in this data, 'traffic' here is partly 'time of day'. Chart: `figures/27_weather_traffic_late_rate.png`.

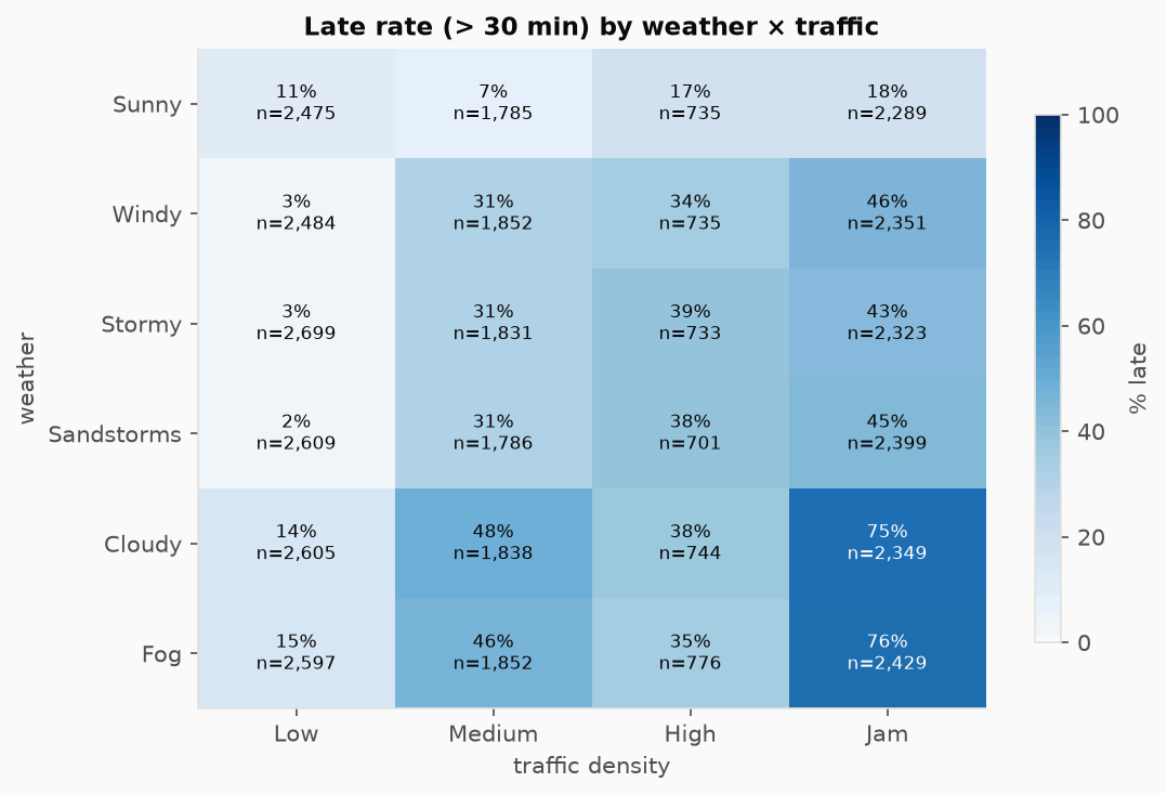


Figure 27 — Weather traffic late rate (source: figures/27_weather_traffic_late_rate.png)

4.4 Distance vs delivery time — how much does distance alone explain? (PDF §8)

A linear model on distance only (fitted on training rows, an additional analysis) explains **R^2 0.087** on the test period (MAE 7.22 min; +0.54 min per km). Squared correlation on all valid rows: 0.103. **Distance is a real but secondary driver:** about one-tenth of the variance.

4.5 Rider experience and rating vs delay (PDF PS3, §8)

Rider age band (experience proxy)	Orders	Mean ETA	% late
20–24	10,782	23.0	17.8
25–29	10,853	23.0	18.1
30–34	10,901	29.6	41.4
35–39	11,112	29.5	41.6

Rider rating band	Orders	Mean ETA	% late
4.5–4.7	17,385	24.2	16.4
4.8–5.0	18,185	24.4	18.5
< 4.5	8,024	35.1	84.6

On similar routes (same distance band and traffic level, 14 strata) [Observed]:

- Orders by riders rated 4.8–5.0, vs riders rated below 4.5: delivery time -8.9 min; late rate -58.9 pp.
- Orders by riders aged 35–39, vs riders aged 20–24: delivery time +6.6 min; late rate +24.2 pp.

Caveats:

- **Rating may be a consequence, not a cause:** a late delivery can lower the rider's rating, which is the leakage risk recorded since Stage 1.
- **There is no tenure field:** age is the only experience proxy, and older riders are slower here, not faster.
- **Rows are orders, not riders** (§9).
- These are dispatch-time fields, so they are excluded from the placement models.

4.6 Three evidence-backed findings (PDF §7)

- **Finding 1: delays concentrate in the evening peak.** [Observed] Orders placed 19:00–21:59 are 49.4–50.6% late, against 29.8% for the network. The 15:00–23:59 window carries 84.5% of all late orders. Time of day is also the models' strongest factor. → Recommendation 3.
- **Finding 2: late orders can be flagged at order placement.** [Observed] Using only order-time information, the classifier reaches ROC-AUC 0.791 on unseen days, catches 81.3% of late orders, and its riskiest 10% of orders were 69.8% late. → Recommendation 1.
- **Finding 3: most ETA uncertainty is resolved only after dispatch.** [Observed] The placement ETA explains 31% of the variance (MAE 6.19 min). A comparison model with dispatch-time data, without rider rating, explains 76% (MAE 3.67 min). Distance alone explains only 9%. → Recommendation 2.

Chart: `figures/25_late_rate_ranges.png` compares how much late rates vary by each factor.

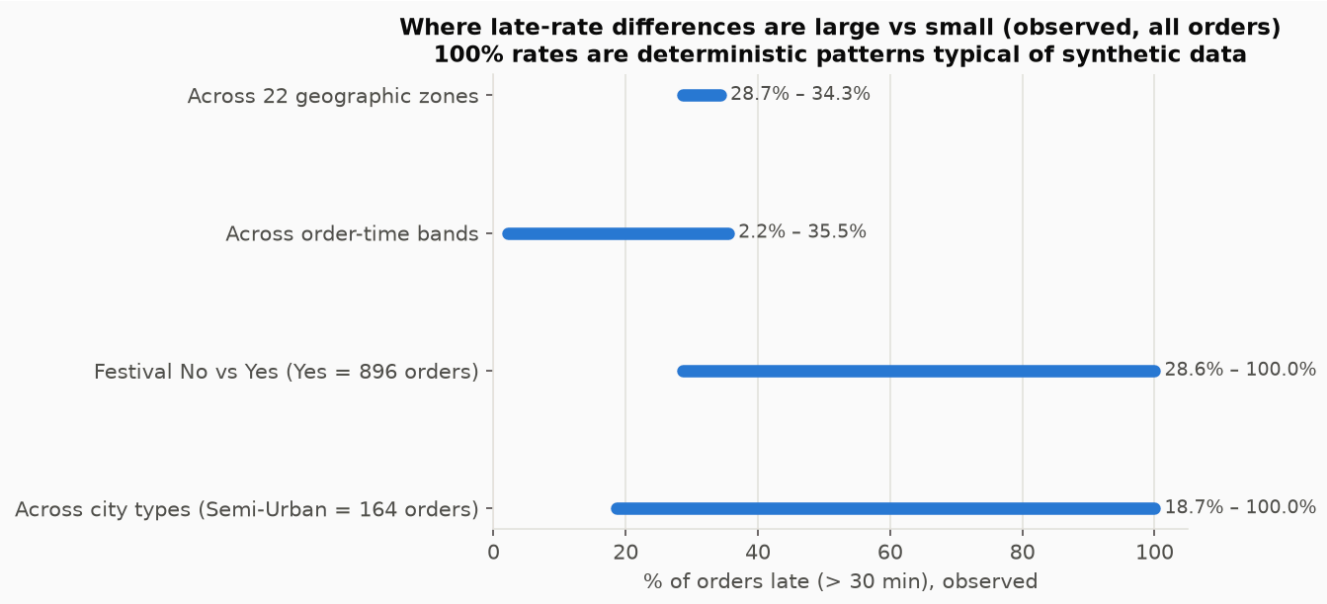


Figure 25 — Late rate ranges (source: figures/25_late_rate_ranges.png)

5. Regression Model Findings (ETA at order placement)

Model	Test MAE (min)	Test RMSE	Test R^2
Median baseline	7.47	9.31	-0.000
Ridge regression	6.42	8.00	0.262
Random forest	6.20	7.76	0.305
Gradient boosting (selected)	6.19	7.73	0.310

- **What MAE 6.19 means:** a typical prediction is off by about 6 minutes on an average delivery of 26 minutes. That is usable for a delivery *window*, not a precise minute promise.
- The model explains 31% of the variation. The rest depends on dispatch-time factors (rider, batching, live conditions) that are deliberately unavailable at placement (§7, Recommendation 2).
- **Errors are largest where deliveries are slowest:** Jam traffic MAE 7.31 vs Low 5.12; 15–25 km 7.04 vs 0–5 km 5.25.
- Model selection, the feature choices (`order_month` dropped; missing-time and missing-festival rows kept) and tuning were all made on training data before the test set was scored (Stage 4).

Charts: `figures/12_regression_model_comparison.png`, `figures/13_eta_pred_vs_actual.png`, `figures/14_eta_error_by_segment.png`.

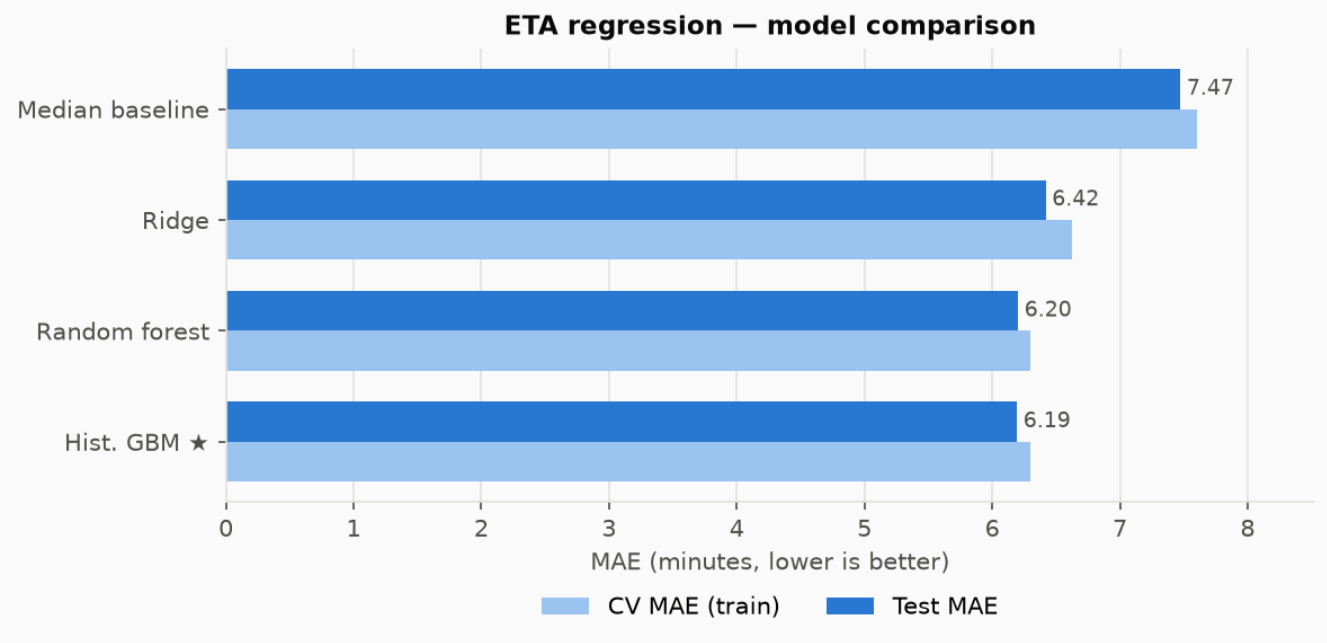


Figure 12 — Regression model comparison (source: figures/12_regression_model_comparison.png)

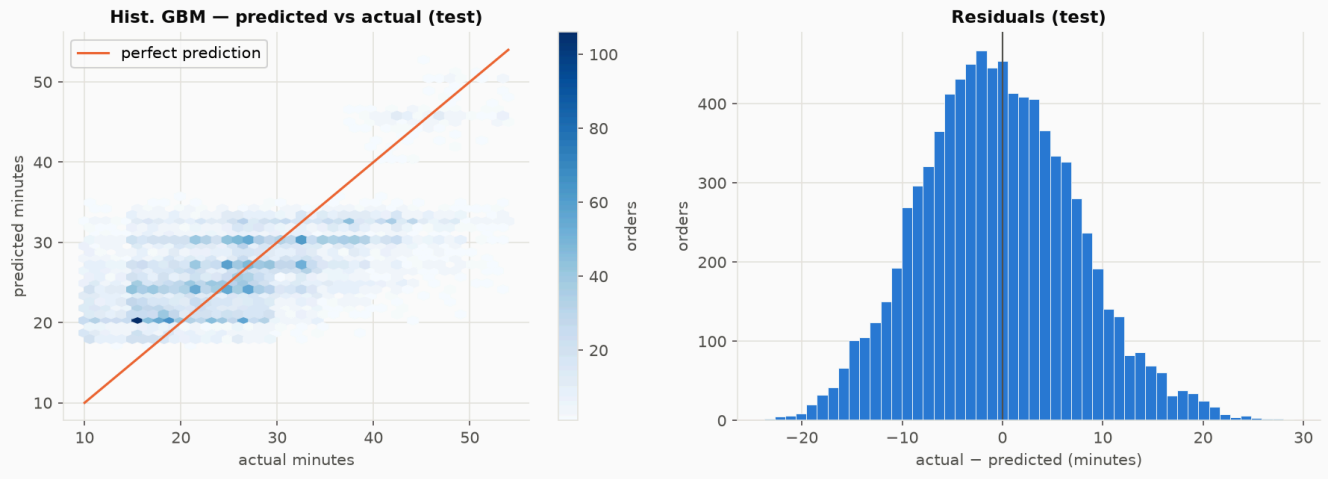


Figure 13 — Eta pred vs actual (source: figures/13_eta_pred_vs_actual.png)

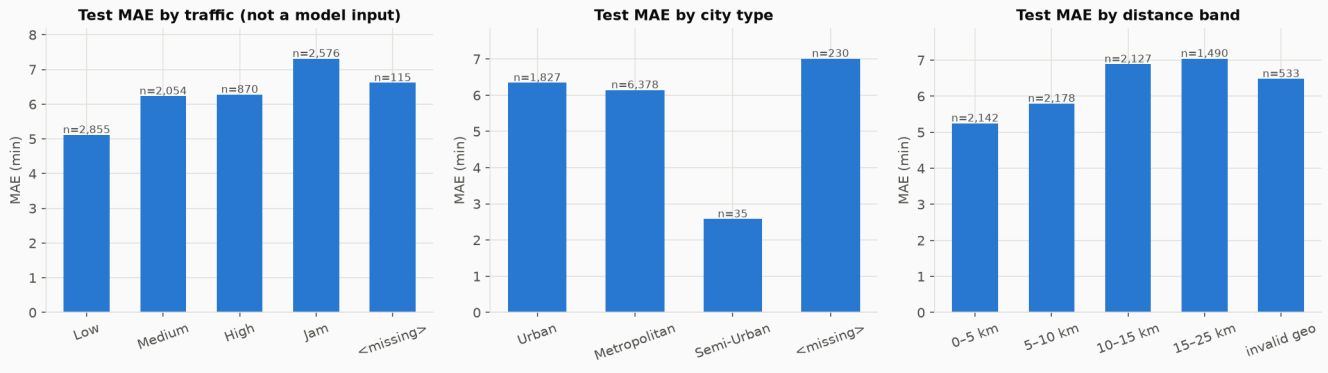


Figure 14 — Eta error by segment (source: figures/14_eta_error_by_segment.png)

5.1 Model selection table (PDF §7)

Task	Model	Selection metric (training CV)	Test result	Role
ETA	Mean baseline (training mean 26.34 min)	—	MAE 7.51 · RMSE 9.31 · R^2 -0.001	Naive baseline named in the PDF
ETA	Median baseline	CV MAE 7.608	MAE 7.47 · R^2 -0.000	Naive baseline (Stage 4)
ETA	Distance-only linear model	—	MAE 7.22 · R^2 0.087	Additional analysis
ETA	Ridge regression	CV MAE 6.621	MAE 6.42 · R^2 0.262	Candidate
ETA	Random forest	CV MAE 6.299	MAE 6.20 · R^2 0.305	Candidate
ETA	**Gradient boosting**	**CV MAE 6.295**	**MAE 6.19 · RMSE 7.73 · R^2 0.310**	**Final**
Late > 30 min	Logistic regression	CV PR-AUC 0.560	ROC-AUC 0.766 · PR-AUC 0.570 · P 0.402 · R 0.900 · F1 0.556	Candidate
Late > 30 min	**Gradient boosting**	**CV PR-AUC 0.595**	**ROC-AUC 0.791 · PR-AUC 0.608 · P 0.467 · R 0.813 · F1 0.593**	**Final**
Late > 30 min	Prior (constant) baseline	CV PR-AUC 0.300	ROC-AUC 0.500 · recall 0	Naive baseline
Zones	**K-Means k = 22**	Silhouette (max over k = 2–30)	Silhouette 0.948	Final (zones = cities, §8)
Rider-attribute profiles	**K-Means k = 3**	Silhouette (max over k = 2–8)	Silhouette 0.458	Order-level alternative (§9)

Reasons for the final choices:

- **Gradient boosting for ETA:** lowest training-CV MAE, and 17.5% lower test MAE than the mean baseline. Random forest was within 0.004 min (a tie); gradient boosting is lighter to serve, so the extra complexity is not a cost here.
- **Gradient boosting for late risk:** highest CV PR-AUC, the right metric when late orders are the minority.
- **K-Means zones:** maximum silhouette.
- Every choice was made on training data before the test set was scored.

6. Late-Order Classification Findings

The 30-minute line is a **project-defined benchmark** (the dataset has no official quick-commerce SLA).

Metric (test period)	Value	Meaning
ROC-AUC	0.791	A random late order is ranked above a random on-time order 79% of the time
PR-AUC	0.608	vs 0.287 for a no-skill model
Recall	0.813	81.3% of late orders are flagged in advance
Precision	0.467	46.7% of flagged orders are actually late; 53.3% are false alarms
F1	0.593	Balance of the two at threshold 0.268
Brier score	0.159	Probabilities are **not calibrated** — use them for ranking

Why recall matters: a late order the model misses gets no proactive message and no dispatch attention, and it is the costliest failure. The threshold was therefore set below 0.5, at the F1-best point on training data, to favour catching late orders.

What the precision means: at this threshold about half of all orders are flagged, and more than half of those alerts are false. That is acceptable only for cheap interventions. Costly actions should be reserved for the highest-risk orders (Recommendation 1).

Per 1,000 orders (test period): 500 flagged → 234 truly late + 267 false alarms; 54 late orders missed.

Risk deciles (test period; reporting only, not used for any model choice):

Risk decile (1 = riskiest)	Orders	% actually late
1	847	69.8
2	847	52.4
3	847	44.5
4	847	34.1
5	847	32.7
6	847	20.0
7	847	15.8
8	847	11.8
9	847	3.5
10	847	2.5

Chart: `figures/24_late_risk_deciles.png`.

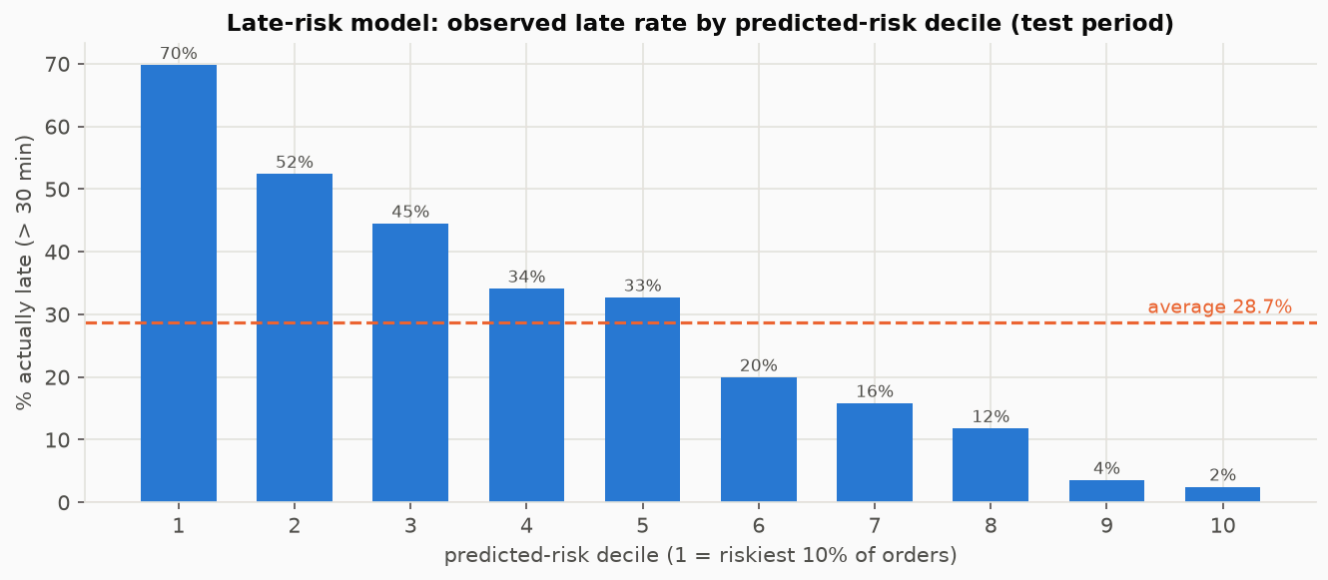


Figure 24 — Late risk deciles (source: figures/24_late_risk_deciles.png)

Benchmark sensitivity:

Benchmark	Late share	Flagged share	Precision	Recall	F1	ROC-AUC	PR-AUC
> 30 min	28.7%	50.0%	0.467	0.813	0.593	0.791	0.608
> 35 min	16.7%	31.4%	0.393	0.738	0.513	0.831	0.540
> 40 min	8.7%	14.9%	0.336	0.577	0.424	0.852	0.472

- **Stricter benchmarks rank better but alert worse.** ROC-AUC rises to 0.852 at > 40 min because very slow orders are more distinctive. But they are rarer (8.7%), so precision falls to 0.336 and recall to 0.577.
- **> 30 min is the most operationally useful benchmark:** it has the highest F1 (0.593 vs 0.513 and 0.424).

Charts: `figures/16_classification_roc_pr.png`, `figures/17_classification_threshold.png`, `figures/18_confusion_matrix.png`, `figures/21_late_threshold_sensitivity.png`.

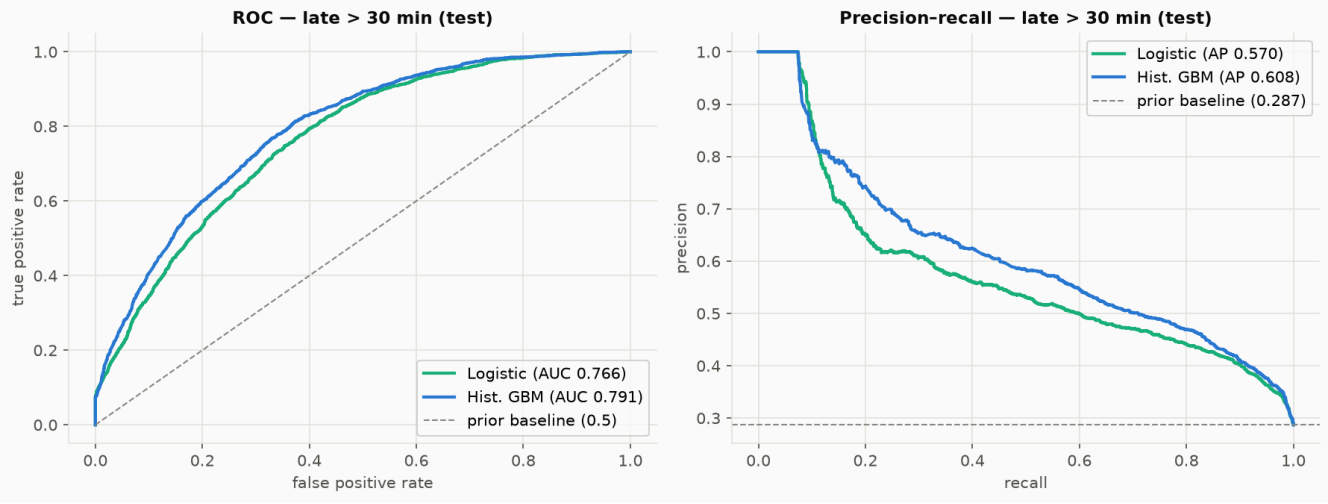


Figure 16 — Classification roc pr (source: figures/16_classification_roc_pr.png)

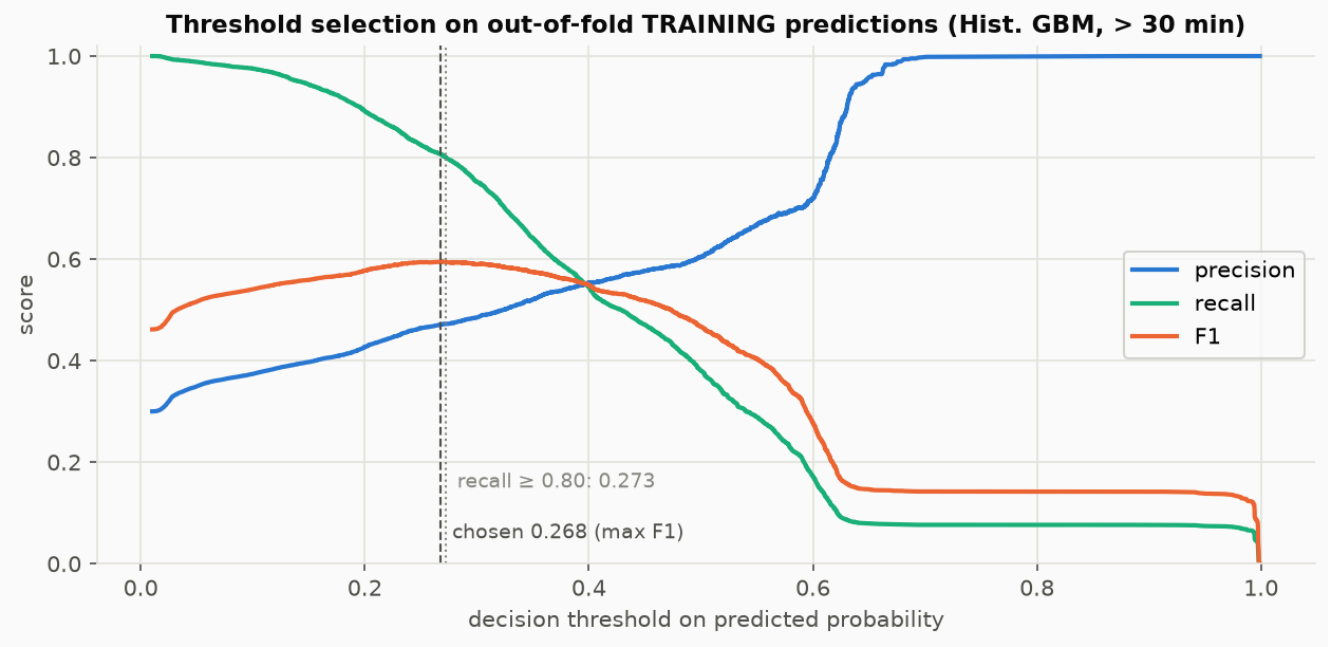


Figure 17 — Classification threshold (source: figures/17_classification_threshold.png)

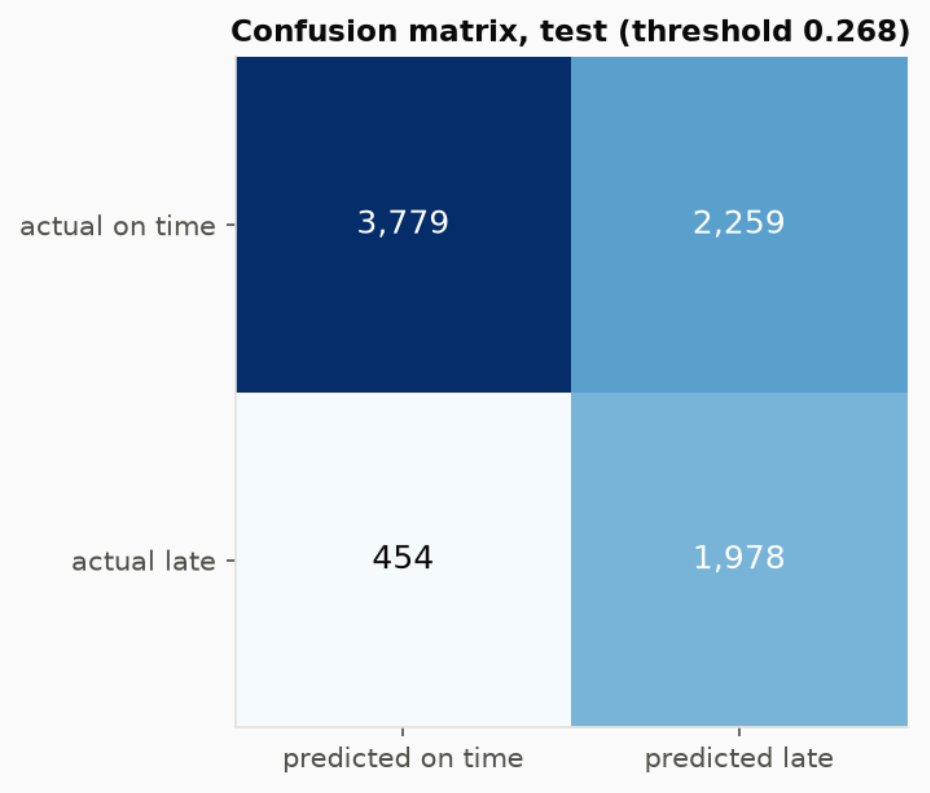


Figure 18 — Confusion matrix (source: figures/18_confusion_matrix.png)

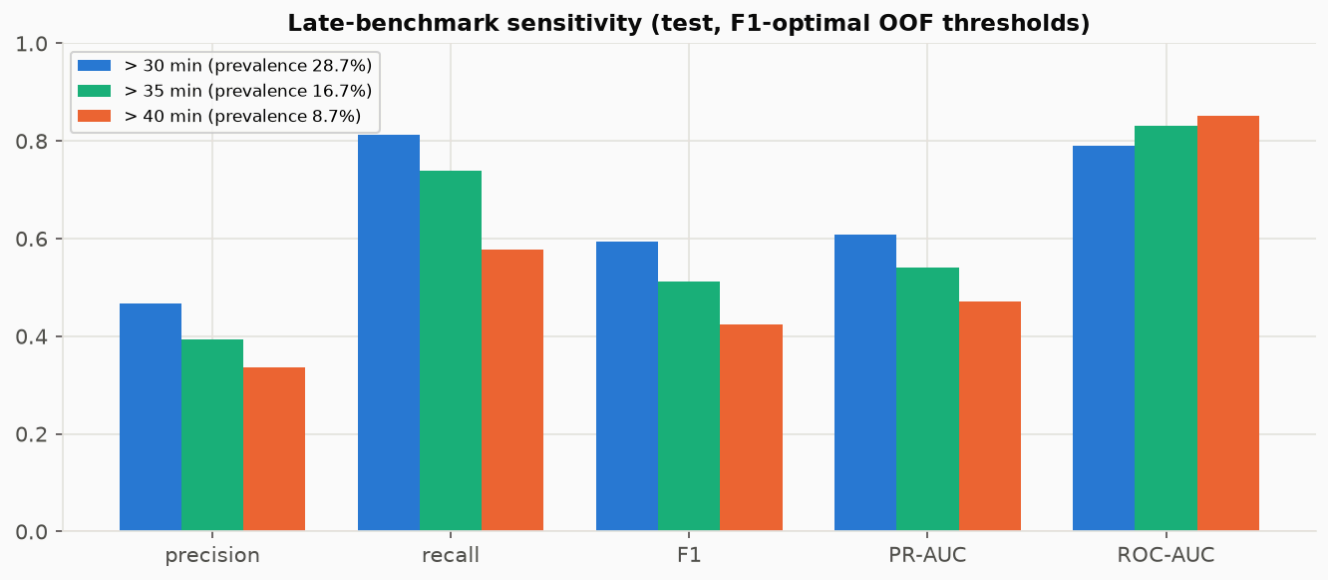


Figure 21 — Late threshold sensitivity (source: figures/21_late_threshold_sensitivity.png)

Class imbalance handling (PDF PS5):

- Late orders are 30.1% of training data and 28.7% of test data.
- Models were selected on PR-AUC, which is sensitive to the minority class, not accuracy.
- The decision threshold was moved from 0.5 to the F1-best point on out-of-fold training predictions.
- No resampling was needed: threshold moving fixes the recall/precision balance directly and keeps probabilities interpretable.

Chosen threshold on the curves: at 0.268 → recall 0.813, precision 0.467, false positive rate 0.374, marked in `figures/29_threshold_on_pr_roc.png`.

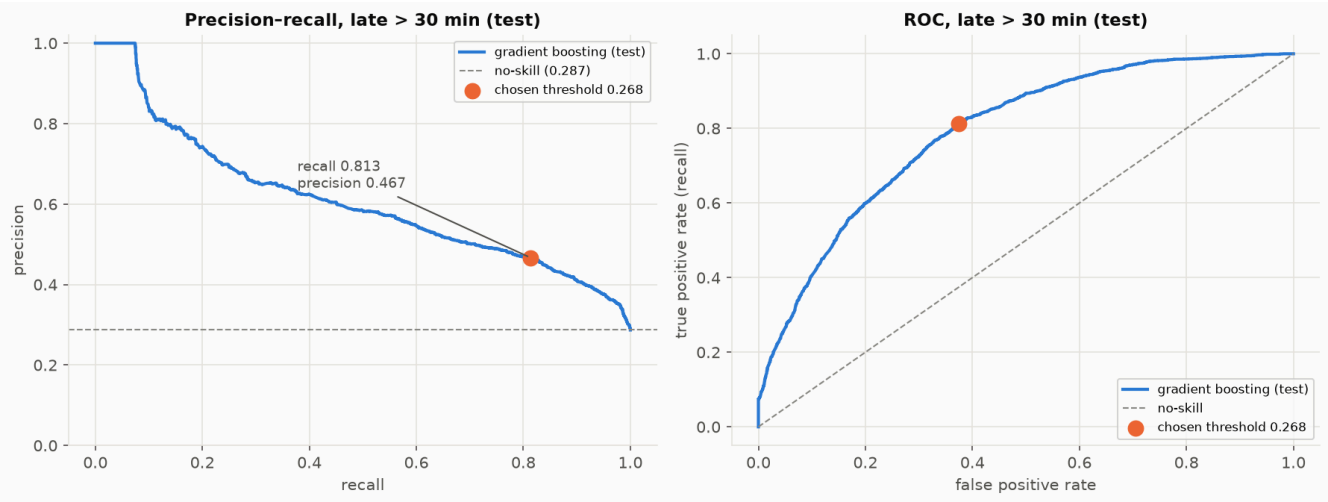


Figure 29 — Threshold on pr roc (source: figures/29_threshold_on_pr_roc.png)

7. Model Explainability

Stage 4 already provided:

- single-feature permutation importance (ETA, test set);
- predicted-vs-actual and residual plots;
- ROC and precision-recall curves;
- model comparison tables.

Stage 5 adds two things for the saved primary models, using only the 13 order-placement features, on **training split only (37,123 rows, 2022-02-11 → 2022-03-29)**, without retraining or test data:

- **grouped** permutation importance for **both** models;
- partial-dependence curves.

Grouping permutes collinear features together, so the four hour features are scored as one 'time of day' effect. SHAP was not used: the package isn't installed, and the brief asks for no unnecessary installs.

Factor	ETA model: training MAE increase when shuffled (min)	Late model: ROC-AUC drop when shuffled
Time of day	1.002 ± 0.021	0.1598
Delivery distance	0.642 ± 0.009	0.0849
Festival day	0.390 ± 0.004	0.0182
City type	0.263 ± 0.008	0.0169
Restaurant location	0.042 ± 0.001	0.0077
Calendar (day of week / month day / weekend)	0.038 ± 0.002	0.0043
Order type	0.010 ± 0.000	0.0017

Baselines on training data: ETA MAE 6.13 min; late-risk ROC-AUC 0.806. Chart: `figures/22_explainability_grouped_importance.png`.

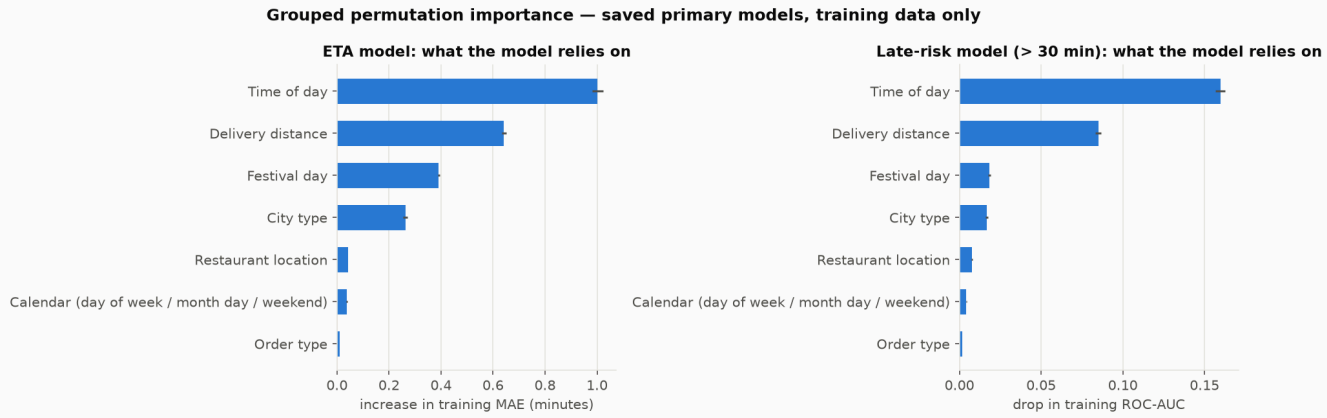


Figure 22 — Explainability grouped importance (source: figures/22_explainability_grouped_importance.png)

In plain language (what the model has learned):

- **Time of day:** the model's average ETA is about 21.8 min for 08–10 orders and 29.1 min at 20:00. The predicted chance of being late rises from 4% at 09:00 to 42% at 20:00.
- **Distance:** predicted ETA rises from 22.7 min at 2 km to 28.5 min at 20 km, mostly as a step beyond about 10 km (24.5 → 28.5 min).
- **Festival day:** 26.1 → 40.2 min.
- **City type:** Urban 24.0, Metropolitan 27.0, Semi-Urban 41.7 min. Semi-Urban has only 164 orders, so it is unstable.
- **Little effect:** restaurant location, day of week or month and order type.

Association, not causation: these show what the model uses to predict, not what *causes* delay. For example, the hour effect may reflect traffic, demand or kitchen load; the data cannot separate them, since traffic follows the hour almost exactly. The test-set importance ranking from Stage 4 agrees: hour and distance are top. Chart: `figures/23_eta_partial_dependence.png`.

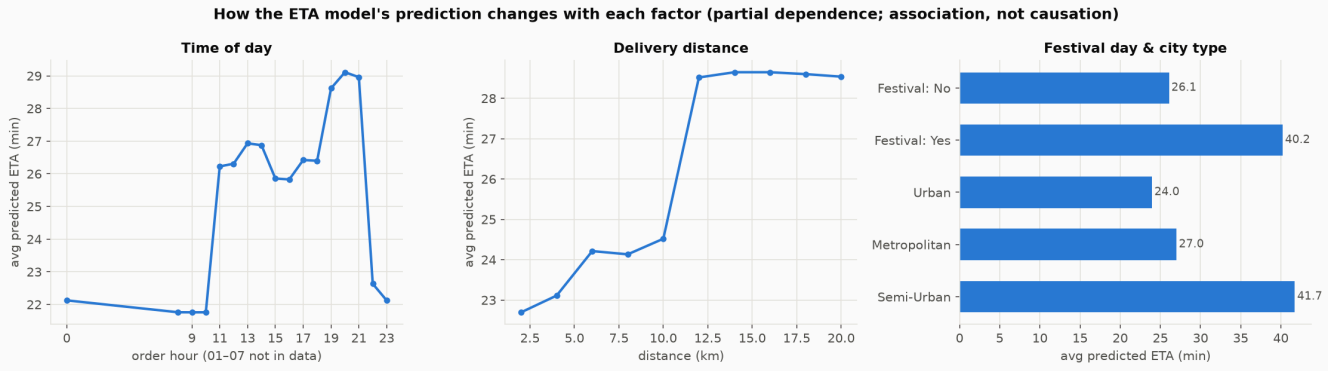


Figure 23 — Eta partial dependence (source: figures/23_eta_partial_dependence.png)

8. Geographic and Dark-Store Analysis

Check	Result
K-Means zones selected (Stage 4)	k = 22, silhouette 0.948
Zones vs cities	Every zone holds exactly one city, and every city one zone: **zones = cities** (yes)
Silhouette of the *city labels* themselves	**0.948** (identical), so the high score is just distance between cities
Late-rate range across zones	28.7% – 34.3%; chi-square p = **0.103** (no significant difference)
Median ETA range across zones	25 – 27 min
Late-rate range across order-time bands (comparison)	2.2% – 35.5%
Zone sizes	478 – 3,443 valid orders
Orders without a valid location	3,640 overall; concentrated in VAD 50.2%, DEH 33.2%, BHP 30.8%
Delivery points	Synthetic diagonal offsets from 388 restaurant locations; median distance 9.2 km (max 21.0)

8.1 Zone vs network benchmark (PDF PS6, §8)

Network benchmark (valid-location orders): mean ETA 26.31 min, late rate 29.94%. High demand = at least the median zone volume (2070 orders). Significance: chi-square (late rate) and Welch t-test (ETA) for each zone vs the rest, Bonferroni-corrected for 22 zones.

Zone	City	Orders	Demand share	Mean ETA	vs network	% late	vs network	Late p	High demand & above benchmark
19	ALH	568	1.4%	27.40	+1.09	34.3%	+4.4 pp	0.024	
16	DEH	492	1.2%	26.64	+0.33	33.5%	+3.6 pp	0.088	
17	AURG	555	1.3%	26.85	+0.54	33.2%	+3.2 pp	0.105	
4	JAP	3,443	8.2%	26.69	+0.38	31.9%	+2.0 pp	0.008	yes
20	BHP	478	1.1%	26.72	+0.41	31.8%	+1.9 pp	0.398	
14	AGR	550	1.3%	26.64	+0.32	31.6%	+1.7 pp	0.407	
9	LUDH	687	1.6%	26.23	-0.08	31.6%	+1.7 pp	0.363	
18	VAD	1,576	3.8%	26.66	+0.35	30.8%	+0.9 pp	0.442	
0	HYD	3,181	7.6%	26.43	+0.12	30.5%	+0.6 pp	0.488	yes
3	RANCHI	2,564	6.1%	26.39	+0.08	30.4%	+0.5 pp	0.595	yes
6	PUNE	3,132	7.5%	26.34	+0.03	29.6%	-0.3 pp	0.712	

Zone	City	Orders	Demand share	Mean ETA	vs network	% late	vs network	Late p	High demand & above benchmark
8	COIMB	3,170	7.6%	26.29	-0.02	29.6%	-0.4 pp	0.644	
10	CHEN	3,145	7.5%	26.36	+0.05	29.4%	-0.5 pp	0.544	
12	KOL	700	1.7%	26.36	+0.05	29.4%	-0.5 pp	0.800	
1	INDO	3,159	7.5%	26.12	-0.20	29.2%	-0.7 pp	0.392	
2	BANG	3,195	7.6%	26.05	-0.26	29.2%	-0.8 pp	0.336	
11	GOA	609	1.5%	26.10	-0.21	29.1%	-0.9 pp	0.668	
5	SUR	3,187	7.6%	26.08	-0.23	29.0%	-0.9 pp	0.251	
15	MUM	3,173	7.6%	26.05	-0.26	29.0%	-0.9 pp	0.236	
7	KNP	676	1.6%	26.16	-0.15	29.0%	-0.9 pp	0.619	
13	MYS	3,012	7.2%	26.02	-0.29	28.7%	-1.2 pp	0.135	
21	KOC	701	1.7%	26.35	+0.03	28.7%	-1.3 pp	0.487	

Result: the PDF's 'high demand and above benchmark' rule flags HYD, RANCHI, JAP. However, **0 zones differ significantly in late rate and 0 in ETA** after correction, so these are **provisional watch-list cities, not dark-store recommendations**. The largest gaps are about 1 minute of ETA, far smaller than the evening-peak effect.

What the analysis can establish: demand is spread across 22 well-separated cities, and the restaurant-to-customer distances in the data. Distance itself matters (§4).

What it cannot establish:

- operational zones inside a city;
- demand hot-spots;
- **any specific dark-store location.**

Customer positions are generated offsets, not real addresses. Zone late rates are statistically indistinguishable. Several cities lose 30–50% of their orders to invalid coordinates. The clusters should be presented as *city groupings*, not validated delivery zones.

Data needed for dark-store placement decisions:

- real customer delivery coordinates with order timestamps, to measure demand density by hour;
- SKU-level demand and the inventory constraints of a dark store;
- candidate site availability, rent and operating costs;
- rider supply and road-network travel times (not straight-line distance);
- a target service radius or promised delivery time.

Charts: `figures/19_kmeans_elbow_silhouette.png`, `figures/20_delivery_zones.png`, `figures/25_late_rate_ranges.png`.

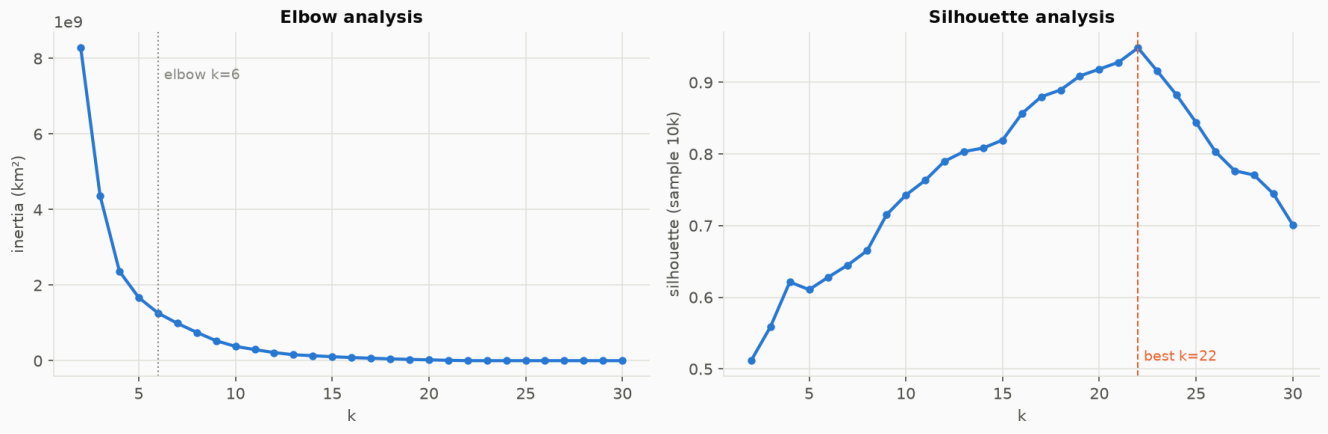


Figure 19 — Kmeans elbow silhouette (source: figures/19_kmeans_elbow_silhouette.png)

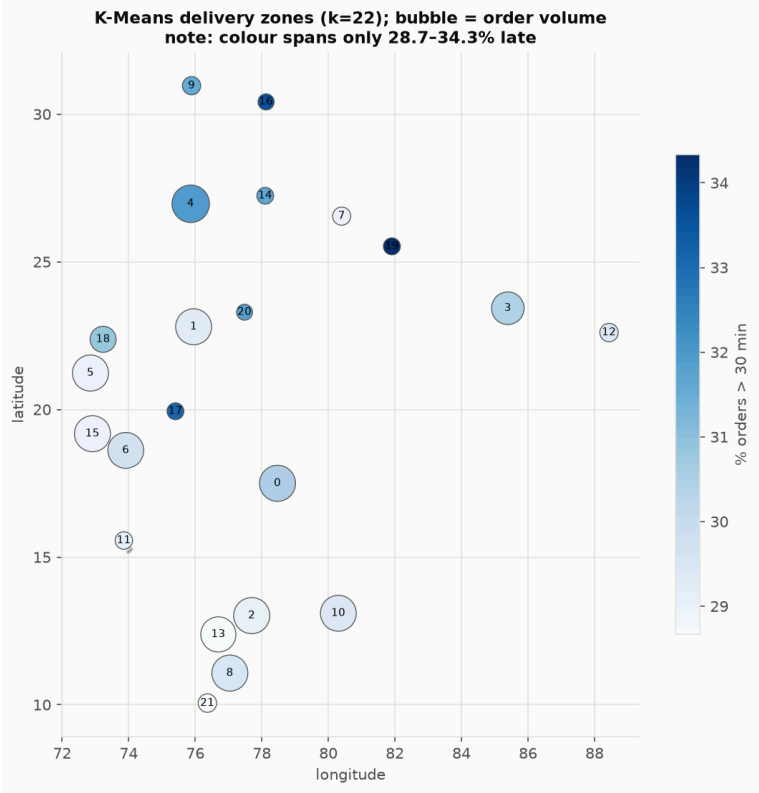


Figure 20 — Delivery zones (source: figures/20_delivery_zones.png)

9. Rider Segmentation Feasibility

Evidence (cleaned data)	Value
Distinct `delivery_person_id` values	1,320
IDs with more than one recorded age	**100%** (ages per ID: 4–22, median 17)
IDs with more than one vehicle type	100%
Valid restaurant locations per ID	at most 1 (each ID is tied to one restaurant)
ID suffixes	DEL01, DEL02, DEL03 only

Verdict: rider segmentation is not feasible.

- The ID pattern `<CITY>RES<nn>DEL<nn>` is a restaurant-plus-slot code: one ID covers many people of different ages.
- Any 'rider segment' built on it would describe restaurants or slots, not riders.

- No rider segments were produced, and no individual rider performance is claimed.

What is possible instead (labelled accurately): an *order-level delivery-condition profile*. This is not rider segmentation; it is dispatch-time information and is confounded.

Vehicle type	Orders	Mean ETA	% late > 30
bicycle	68	26.4	30.9
electric_scooter	3,814	24.5	23.4
motorcycle	26,435	27.6	33.9
scooter	15,276	24.5	24.3

Motorcycles look slower, but vehicle condition 0 occurs only on motorcycles and has a 41.7% late rate, so the two cannot be separated.

9.1 Alternative: order-level rider-attribute profiles (NOT rider segmentation)

PS7 asks for rider segments. Because rider identity is unreliable, K-Means was run on **orders** described by the assigned rider's age and rating (43,594 orders with both values; standardised). k = 3 was chosen by maximum silhouette (0.458 over k = 2–8). Each profile describes orders, not individual riders.

Profile (descriptive name)	Orders	Mean age	Mean rating	Mean ETA	% late	Mean batched deliveries	% motorcycle
Younger, higher-rated	19,948	24.5	4.74	22.1	13.3	0.65	57.4
Older, higher-rated	16,962	34.6	4.75	27.6	27.0	0.78	57.2
Older, lower-rated	6,684	32.0	4.05	35.3	86.4	0.94	64.8

How to use it:

- It is a **dispatch-time risk signal**: orders assigned to the lower-rated profile run late far more often.
- It **cannot** support rostering or training decisions for individual riders. The same 'rider ID' covers many people, and a low rating may *result* from late deliveries.
- A real rider segmentation needs the data listed below.

Chart: `figures/28_rider_attribute_profiles.png` (rating, age as the experience proxy, and late rate, as the PDF's visual set asks).

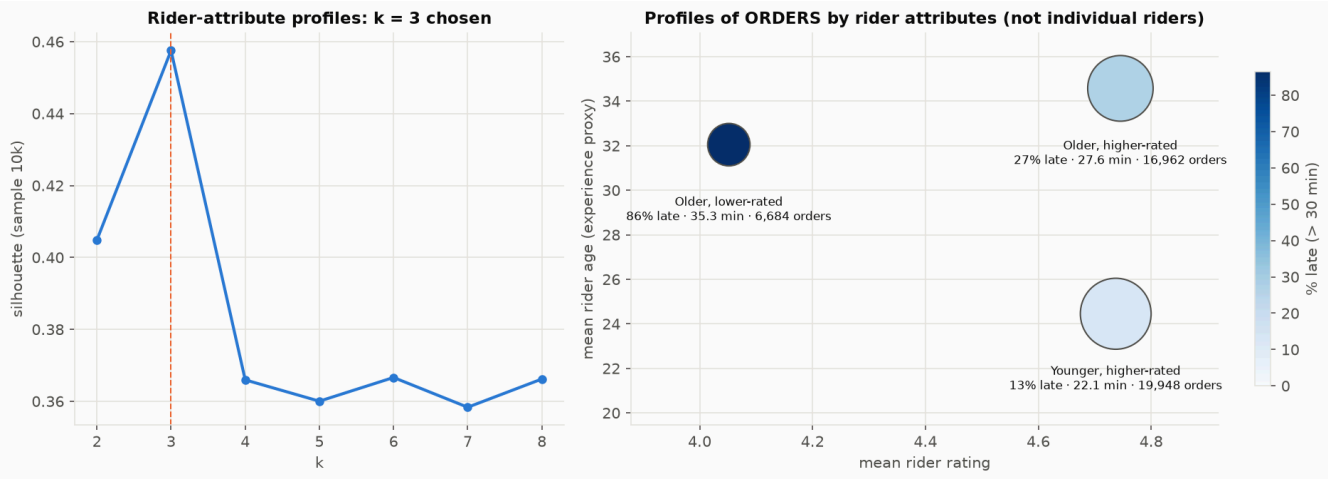


Figure 28 — Rider attribute profiles (source: figures/28_rider_attribute_profiles.png)

Data needed for genuine rider segmentation:

- a stable, unique rider identifier;
- per-rider trip history across restaurants;
- shift and login times;
- assignment, pickup and drop timestamps;
- the rider's own vehicle, and ratings captured before each order.

10. Exactly Three Prioritised Recommendations

Full detail (evidence, implementation, KPIs, risks): `reports/business_recommendations.md`.

Priority	Recommendation	Core evidence [Observed]
1	Deploy order-placement late-risk alerts with tiered interventions	Recall 81.3%, ROC-AUC 0.791; riskiest 10% → 69.8% late
2	Introduce a two-stage ETA: estimate at placement, refresh at rider assignment	Placement MAE 6.19 → 3.67 min with dispatch-time data (comparison model, no rating)
3	Surge-staff riders in the 19:00–21:59 window and plan promises around predictable peaks	15:00–23:59 = 72.9% of orders, 84.5% of late orders

Why this order:

- **#1** uses a model that is already trained, saved and validated for order-placement use.
- **#2** has the largest measured accuracy gain, but needs a field-timing audit and a new model first.
- **#3** is a planning change supported by strong, consistent patterns. Their cause is not established.

A dark-store or zone recommendation was considered and **rejected**: the evidence does not support it (§8).

11. Expected Business Value and KPIs

Recommendation	Measured in this project	Hypothetical scenario (not measured)	Primary KPI
1. Late-risk alerts	81.3% of late orders flagged before they are late	Late rate −2.3 to −7.0 pp if alerts prevent 10–30% of the late orders they catch	Late rate, pilot vs control; alert precision and recall
2. Two-stage ETA	MAE 3.67 vs 6.19 min (comparison model)	Fewer status-enquiry contacts	ETA MAE at assignment; % within ±5 min
3. Peak capacity planning	Peak window = 84.5% of late orders	Late rate −1.5 pp if the evening late rate matched the afternoon	Late rate by hour band and on festival days

Scenario calculations are transparent arithmetic on observed rates (see the recommendations file). They are not forecasts and must be replaced by pilot measurements.

12. Limitations and Deployment Considerations

Limitation	Effect on deployment and generalisation
Food-delivery data, not genuine quick-commerce grocery delivery	Distances and times are much longer than quick-commerce operations. Re-train on real quick-commerce data before use.
Synthetic or artificial patterns (diagonal delivery offsets, 5-minute target blocks, traffic set by hour, segments 100% late)	Metrics may be optimistic and some effects may be artifacts. Validate on real data.
Test period covers only 8 days (2022-03-30 → 2022-04-06)	No seasonality or longer-term drift is tested. Monitor performance weekly after launch.

Limitation	Effect on deployment and generalisation
April appears only in the test period	The month feature was dropped. Behaviour in unseen months is unverified.
3,640 orders lack valid coordinates (up to 50% in some cities)	Distance is unknown for them (filled with the median at model time), and geographic conclusions under-represent those cities.
1,731 orders lack an order time	Their delivery times look normal (p = 0.78), but their hour is unknown and their errors are higher.
Some values corrected by documented rules (431 sign fixes, 91 placeholders blanked)	The corrections are reversible and audited (`*_raw`, `dq_*` columns).
The > 30-minute late line is project-defined, not an official SLA	Re-set the threshold to the operator's real promise; sensitivity at 35/40 min is provided.
Classification probabilities are not calibrated	Use them for ranking and tiering. Calibrate before showing a probability to customers.
Clustering does not establish dark-store locations	Do not use the zones for site selection (§8).
Rider segmentation is not reliable with current IDs	No rider-level decisions should be based on this data (§9).
Dispatch-time comparison model is unsaved and may leak (rider rating)	Supports Recommendation 2's direction only; it is not a deployable model.

Deployment guidance:

- Use only the saved order-placement models (13 features).
- Run them in shadow mode before acting on them.
- Recalibrate the threshold and probabilities on live data.
- Retrain on a schedule, re-checking the chronological split and the leakage guard.

13. Conclusion

With order-time information alone, DeliveryIQ beats a naive ETA by 17.1% and flags 81.3% of late orders before they happen. That supports an alerting workflow now (Recommendation 1) and capacity planning around predictable peaks (Recommendation 3).

The largest accuracy gain needs information available only after dispatch (Recommendation 2). The geographic and rider analyses were taken as far as the data honestly allows: the clusters are cities, and rider IDs are not people.

The next step is Stage 6: an interactive demo of the placement-time ETA and late-risk models.

Appendix A — Verification of Stage 4 figures

Figure	Value in brief	Recomputed from saved models / metrics	Match
eta_MAE	6.19	6.1940	✅
eta_RMSE	7.73	7.7335	✅
eta_R2	0.31	0.3101	✅
baseline_MAE	7.47	7.4737	✅
improvement_pct	17.1	17.1227	✅
ROC_AUC	0.791	0.7911	✅
PR_AUC	0.608	0.6084	✅
precision	0.467	0.4668	✅
recall	0.813	0.8133	✅

Figure	Value in brief	Recomputed from saved models / metrics	Match
F1	0.593	0.5932	✅
threshold	0.268	0.2681	✅
dispatch_MAE	3.5	3.4982	✅
dispatch_ROC_AUC	0.98	0.9825	✅
k	22	22.0000	✅
silhouette	0.948	0.9482	✅

Appendix B — Stage 5 validation checks

Check	Result	Detail
input datasets match the recorded SHA-256 hashes	✅ pass	
saved models use only primary order-placement features (strict leakage guard)	✅ pass	13 features
every Stage 4 figure in the brief matches recomputation from saved models / metrics (to stated precision)	✅ pass	mismatches: none
saved models reproduce stored Stage 4 test metrics exactly	✅ pass	
K-Means zones recomputed from saved model reproduce Stage 4 silhouette (±0.005)	✅ pass	0.9483 vs 0.9482
rider-ID evidence recomputed from cleaned data (every ID has >1 age)	✅ pass	
raw, cleaned and engineered datasets unchanged after Stage 5	✅ pass	
explainability computed on training rows only	✅ pass	37,123 rows
exactly three prioritised recommendations	✅ pass	3
every figure referenced in the Stage 5 documents exists	✅ pass	all present
challenge guide PDF found and used for the compliance check	✅ pass	['Hackathon_Challenge_Guide.pdf']
Business Report within the guide's 4–6 pages (estimated 1,800–3,300 words)	✅ pass	2,123 words
Business Solution Document at least 5 pages (estimated ≥ 2,500 words)	✅ pass	6,084 words
Stage 5 documents are complete (no stub markers; every document > 3,000 characters)	✅ pass	stub: none; short files: none
rider profiles labelled as order-level, not rider segmentation	✅ pass	
30-minute benchmark never described as an official SLA	✅ pass	

Appendix C — Charts added in Stage 5

- `figures/22_explainability_grouped_importance.png`
- `figures/23_eta_partial_dependence.png`

- `figures/24_late_risk_deciles.png`
- `figures/25_late_rate_ranges.png`
- `figures/26_late_rate_by_hour.png`
- `figures/27_weather_traffic_late_rate.png`
- `figures/28_rider_attribute_profiles.png`
- `figures/29_threshold_on_pr_roc.png`

Reproduce: `.venv\Scripts\python.exe -m src.business_insights`